9T SRAM

Introduction

The design of Static Random Access Memory (SRAM) cells is crucial in modern low-power and high-performance CMOS applications. Conventional 6-transistor (6T) SRAM cells are widely used, but they suffer from issues such as reduced stability, high power consumption, and sensitivity to process variations at deep submicron technology nodes. To overcome these challenges, a 9-transistor (9T) SRAM cell has been designed using Microwind 3.1 in 90nm CMOS technology.

The 9T configuration offers improved read stability, reduced dynamic power consumption, and enhanced noise margin. Although Static Noise Margin (SNM) and leakage power analyses were not performed due to software limitations, the results demonstrate that the proposed 9T SRAM cell architecture provides a more power-efficient and reliable alternative to the traditional 6T design.

The proposed 9T SRAM cell introduces three additional transistors compared to the conventional 6T cell. These extra transistors help isolate the read path from the storage nodes, thus improving read stability and minimizing voltage disturbances during read operations.

Moreover, the 9T configuration allows better control of leakage currents and enhanced noise immunity, making it suitable for low-voltage applications.

In the Microwind 3.1 layout, the design utilizes a 90nm CMOS process with supply voltages Vdd = 1V. Nodes **Q** and $\bar{\mathbf{Q}}$ store the complementary logic states, while separate control transistors manage read and write operations independently. This separation ensures that the read operation does not disturb the stored data, a key advantage over the conventional 6T design.

Proposed 9T SRAM Cell

A novel 9T SRAM cell with enhanced data stability and reduced leakage power consumption is implemented using **90 nm CMOS technology** in Microwind 3.1. The structure of the proposed cell can be divided into two subcircuits — an **upper 6T memory cell** for data storage and a **lower 3T read access network** for stable read operations.

The **upper subcircuit** functions as a conventional 6T SRAM cell, composed of two cross-coupled inverters (formed by transistors N1, N2, P1, and P2) and two access transistors (N3 and N4) that are controlled by the **write control signal (WR)**. This portion is responsible for storing the data.

The **lower subcircuit** consists of two bitline transistors (N5 and N6) and a single **read access transistor (N7)**. The operations of N5 and N6 depend on the data stored in the memory nodes, while N7 is activated by a separate **read control signal (RD)**.

Write Operation

During a write cycle, the WR signal goes high and the RD signal remains low. The read transistor (N7) is turned off to isolate the read path. The write access transistors (N3 and N4) connect the internal nodes to the bitlines (BL and BLB), allowing data to be written into the storage nodes. For example, when writing a logical ‘0’ at Node1, BL is discharged while BLB is charged, forcing a ‘0’ through N3. The complementary behavior occurs when writing to Node2 through N4.

Read Operation

During a read cycle, the RD signal is asserted high while WR is kept low. The read access transistor (N7) turns on, connecting the storage cell to the read path. Depending on the stored data, either BL or BLB is discharged through N5–N7 or N6–N7, respectively. Since N3 and N4 remain off during this phase, the storage nodes (Node1 and Node2) are **completely isolated from the bitlines**, preventing any voltage disturbance on the stored data.

This separation between the read and write paths ensures **improved read stability** and eliminates read disturb issues common in conventional 6T SRAM designs. The voltage at the node storing a logical

‘0’ remains strictly at ground level during a read, which enhances data retention and minimizes soft errors.

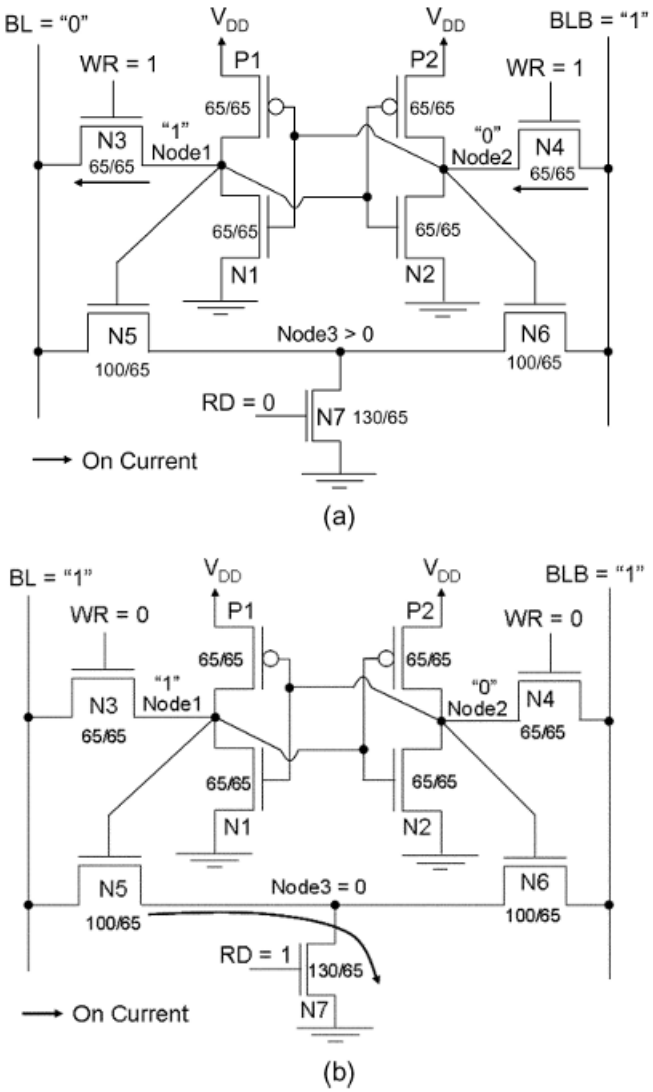
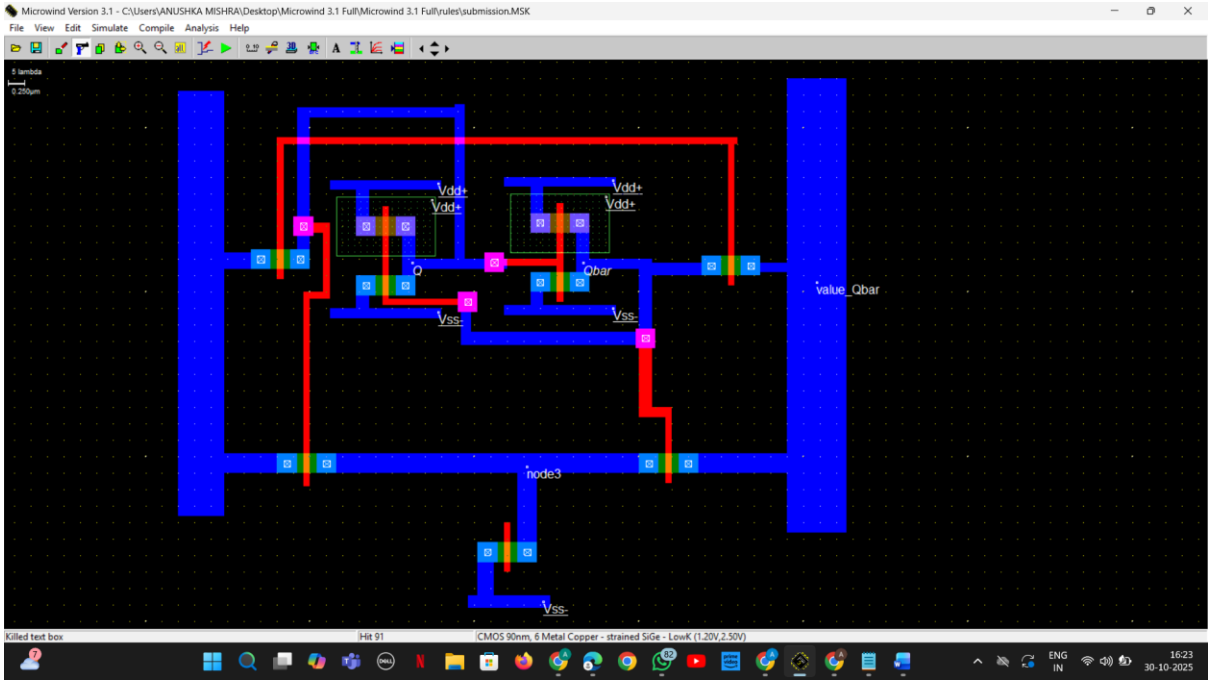
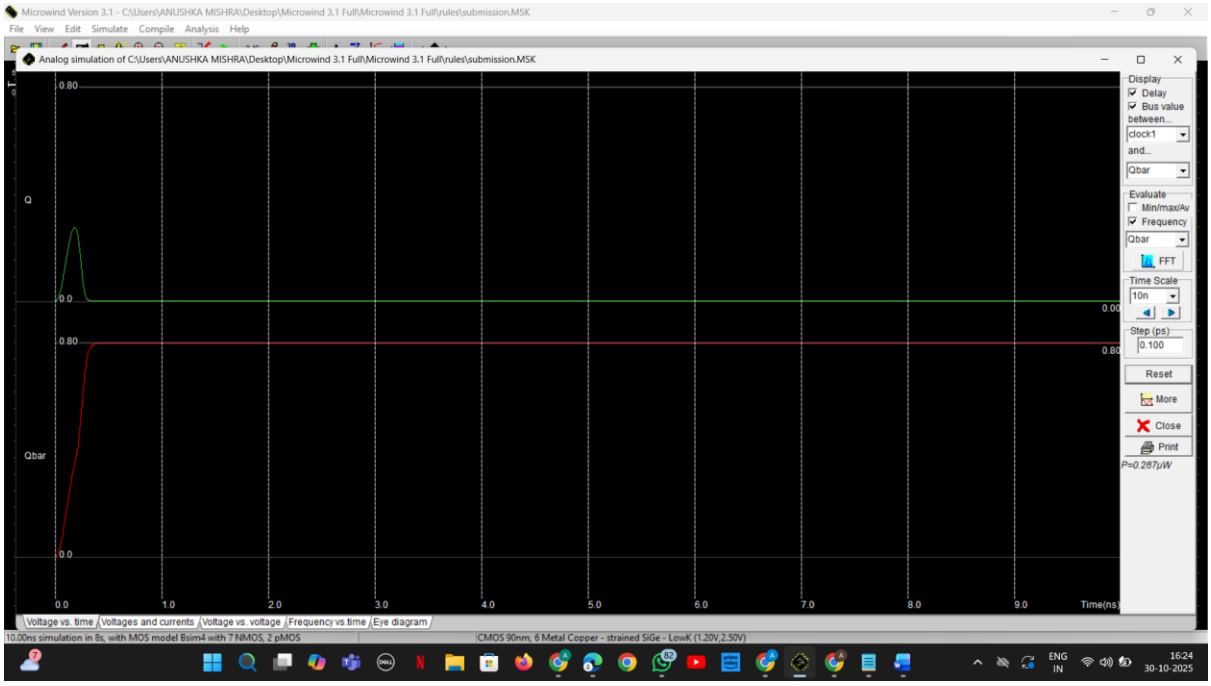


Fig. 2. 9T SRAM cell in the active mode. (a) The 9T SRAM cell during a write operation. (b) The 9T SRAM cell during a read operation. The transistors are sized for a 65-nm CMOS technology.

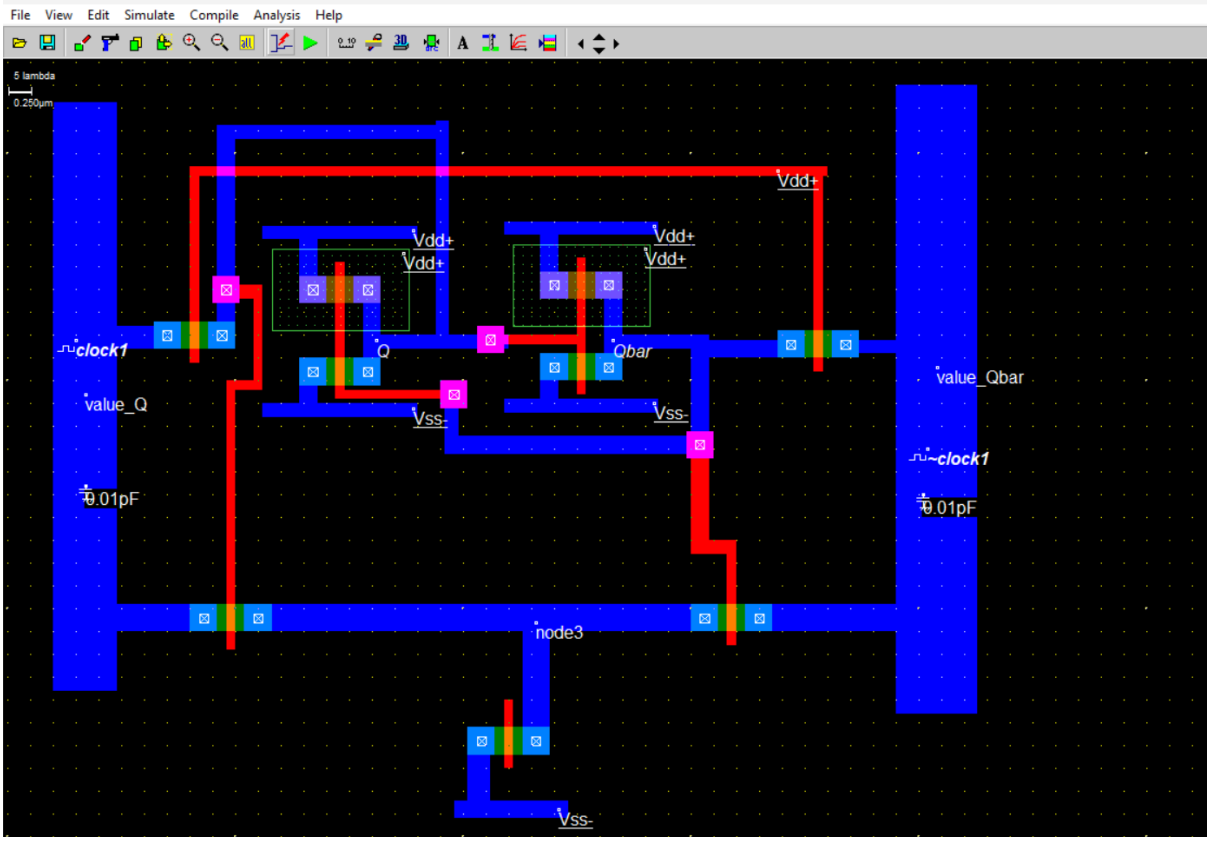
Proposed 9T Sram in static mode



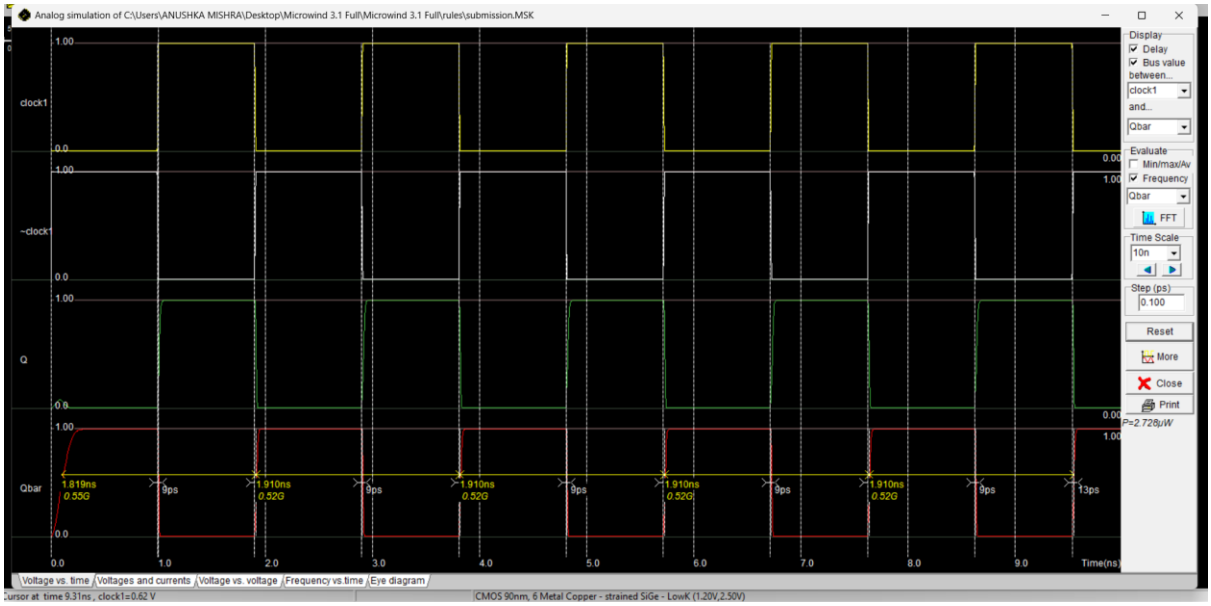
Output



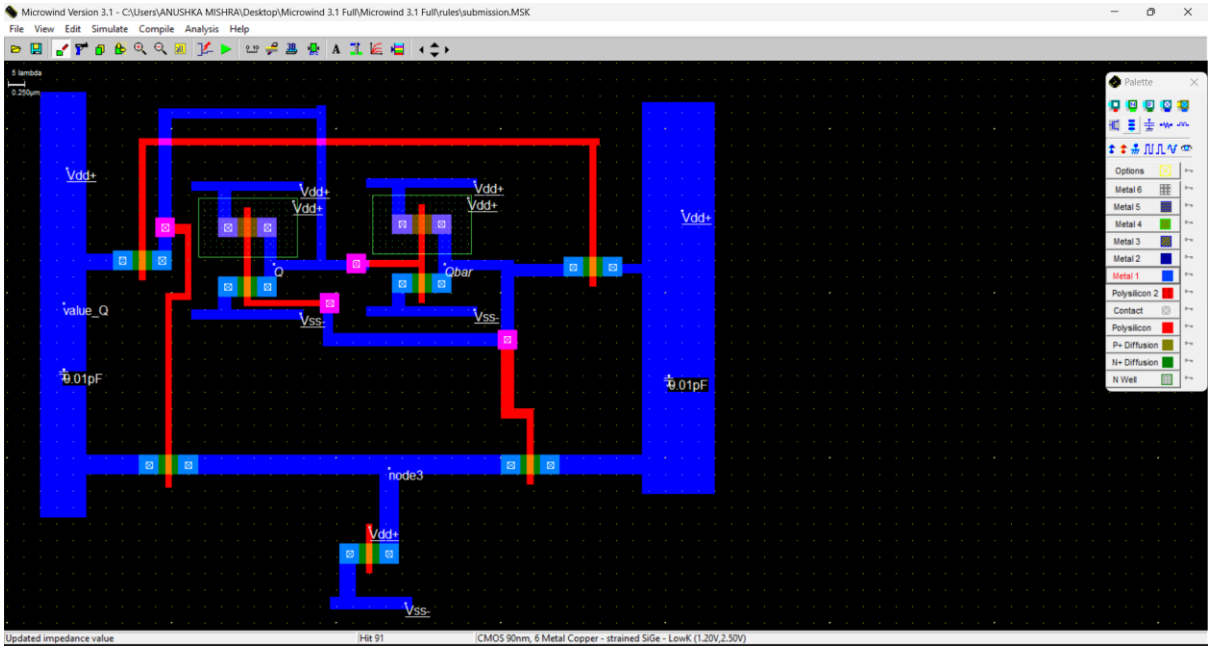
Write Mode



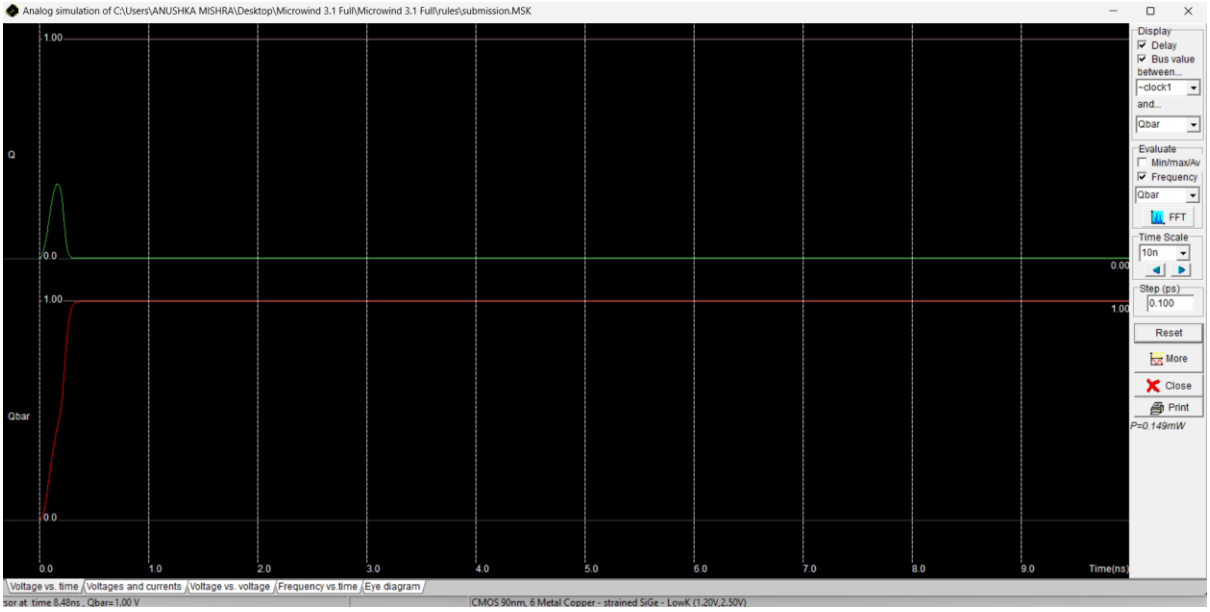
Output



Read Mode



Output



Netlist

CIRCUIT C:\Users\ANUSHKA MISHRA\Desktop\Microwind 3.1 Full\Microwind 3.1 Full\rules\submission.MSK

*

* IC Technology: CMOS 90nm, 6 Metal Copper - strained SiGe - LowK

*

VDD 1 0 DC 1.00

V~clock1 8 0 DC 0 PULSE(1.00 0.00 0.99N 0.01N 0.01N 0.90N 1.91N)

Vclock1 13 0 DC 0 PULSE(0.00 1.00 0.99N 0.01N 0.01N 0.90N 1.91N)

*

* List of nodes

* "Qbar" corresponds to n°4

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* "Q" corresponds to n°6
* "~clock1" corresponds to n°8
* "node3" corresponds to n°9
* "clock1" corresponds to n°13
*
* MOS devices
MN1 0 4 6 0 N1  W= 0.30U L= 0.10U
MN2 0 15 9 0 N1  W= 0.30U L= 0.10U
MN3 0 6 4 0 N1  W= 0.30U L= 0.10U
MN4 9 4 8 0 N1  W= 0.30U L= 0.10U
MN5 4 1 8 0 N1  W= 0.30U L= 0.10U
MN6 13 6 9 0 N1  W= 0.30U L= 0.10U
MN7 13 1 6 0 N1  W= 0.30U L= 0.10U
MP1 1 6 4 1 P1  W= 0.30U L= 0.10U
MP2 1 4 6 1 P1  W= 0.30U L= 0.10U
*
C2 1 0  0.338fF
C3 1 0  0.338fF
C4 4 0  1.320fF
C5 1 0  0.374fF
C6 6 0  1.512fF
C7 1 0  0.374fF
C8 8 0  1.332fF
C9 9 0  0.992fF
C13 13 0  1.281fF
C14 1 0  0.090fF
C15 15 0  0.006fF
```

```
*
* Extra RLC
*
Cadd1 8 0 0.01pF
Cadd2 13 0 0.01pF
*
*
* n-MOS BSIM4 :
* low leakage
.MODEL N1 NMOS LEVEL=14 VTHO=0.28 U0=0.060 TOXE=
1.2E-9 LINT=0.015U
+K1 =0.450 K2=0.100 DVT0=2.300
+DVT1=0.570 LPE0=23.000e-9 ETA0=0.080
+NFACTOR=  0.9 U0=0.060 UA=3.400e-15
+WINT=0.005U LPE0=23.000e-9
+KT1=-0.060 UTE=-1.800 VOFF=0.010
+XJ=0.150U NDEP=170.000e15 PCLM=1.100
+CGSO=100.0p CGDO=100.0p
+CGBO= 60.0p
*
* p-MOS BSIM4:
* low leakage
.MODEL P1 PMOS LEVEL=14 VTHO=-0.32 U0=0.027 TOXE=
1.2E-9 LINT=0.015U
+K1 =0.450 K2=0.100 DVT0=2.300
+DVT1=0.570 LPE0=23.000e-9 ETA0=0.080
+NFACTOR=  1.9 U0=0.027 UA=2.200e-15
+WINT=0.005U LPE0=23.000e-9
```

```
+KT1=-0.060 UTE=-1.800 VOFF=0.010
+XJ=0.150U NDEP=170.000e15 PCLM=0.700
+CGSO=100.0p CGDO=100.0p
+CGBO= 60.0p
*
* Transient analysis
*
* (Winspice)
.options temp=27.0
.control
tran 0.1N 10.00N
print  V(4) V(6) V(8) V(13) > out.txt
plot  V(4) V(6) V(8) V(13)
.endc
.END
```

Conclusion

The design and analysis of a 9T SRAM cell in 90nm CMOS technology show significant advantages over the traditional 6T cell in terms of read stability and dynamic power reduction.

Although SNM and leakage power were not measured, theoretical and architectural improvements indicate that the 9T SRAM offers higher reliability and robustness for low-power, high-performance applications.

This work highlights the potential of 9T SRAM architecture in future VLSI memory design where energy efficiency and stability are critical.